

Indian ETF Quantitative Engine & Time Series Platform

From Static Modern Portfolio Theory to Multi-Asset, Rolling Econometrics & Institutional Risk Management

UNIVERSE EXPANSION

8+ Curated ETFs

Midcap, Nasdaq, Silver, G-Sec

TIME SERIES MODELING

Rolling Sharpe & Vol

EWMA Volatility Clustering

STRESS TESTING

3 Crisis Windows

COVID, 2022 Hikes, 2024 Shock

ARCHITECTURE

100% Automated

Vercel Serverless + PDF Engine

01 FOUNDATION

Multi-Asset Universe

Eliminate Redundancy & Add Alpha

- **Add MID150BEES**
Captures high-growth Indian midcap alpha beyond large-caps.
- **Add MON100 (Nasdaq)**
Global tech exposure + natural USD currency depreciation hedge.
- **Add SILVERBEES**
Dual commodity hedge alongside Gold; industrial demand beta.
- **Add GSEC10IETF**
10-Yr Sovereign G-Sec; real bond yield vs low-return LiquidBees.
- **Split Healing Engine**
Auto-interpolates 1:10 split artifacts across all historical series.

Tech: yfinance, pandas, split-healer

02 ECONOMETRICS

Time Series Modeling

Dynamic Risk & Regime Shifts

- **Rolling Sharpe & Sortino**
180d & 365d rolling windows reveal regime-dependent decay.
- **Rolling Volatility Curve**
Visualizes volatility divergence between equities & gold.
- **EWMA Volatility Clustering**
RiskMetrics decay ($\lambda=0.94$) tracks dynamic variance shocks.
- **Dynamic Beta Tracking**
Measures shifting correlation of Midcap & Bank vs Nifty 50.
- **Correlation Heatmap Matrix**
Pairwise 8x8 matrix exposing low/negative asset linkages.

Tech: scipy.stats, numpy, EWMA engine

03 VALIDATION

Backtests & Stress Tests

Real-World Performance Proof

- **Walk-Forward Rebalancing**
Quarterly & annual rebalancing vs static buy-and-hold.
- **COVID-19 Crash Test**
Stress testing drawdown & recovery during Feb-Apr 2020.
- **2022 Inflation Shock**
Performance through RBI/Fed rate hikes & tech correction.
- **2024 Election Volatility**
Analyzing multi-asset cushion against political gaps.
- **Tail Risk & CVaR (95%)**
Quantifies conditional value at risk during the worst 5% days.

Tech: SLSQP, Dirichlet Monte Carlo

04 DEPLOYMENT

Web Simulator & Reports

Interactive Tool & PDF Export

- **Custom Preset Baskets**
Toggle Classic 5, All-Weather 7, or Pure Equity Alpha.
- **Head-to-Head Compare UI**
Side-by-side radar metrics comparing any two ETFs directly.
- **Interactive Stress Simulator**
Sliders to test portfolio survival in a 20% equity crash.
- **Institutional 12-Page PDF**
Auto-compiled executive brief with clean print typography.
- **25+ Pytest Test Suite**
Full test coverage ensuring zero lookahead bias.

Tech: Vanilla JS, Chart.js, Vercel Serverless

EXECUTION TIMELINE & MILESTONES

Estimated Duration: 3-4 Focused Sprints

Sprint 1

Universe Expansion & Ingestion
Days 1-3

Sprint 2

Time Series & Rolling Econometrics
Days 4-7

Sprint 3

Backtesting & Stress Testing
Days 8-11

Sprint 4

Web UI Simulator & PDF Service
Days 12-14

ARCHITECTURAL BLUEPRINT & UNIVERSE COMPARISON

Solving Correlation Redundancy, Cash Drag & Model Fragility with Modern Multi-Asset Quant Engineering

ETF BASKET UPGRADE: v1.0 (CLASSIC) vs v2.0 (ALL-WEATHER)

Strategic shift from bank-heavy equity to true non-correlated multi-asset allocation

TICKER	ASSET CLASS	ROLE IN PORTFOLIO	v1.0 ISSUE / v2.0 BENEFIT
NIFTYBEES.NS	Large-Cap Equity	Core India Growth Anchor	Baseline market return; high liq
JUNIORBEES.NS	Large/Mid Next 50	Growth & Beta Booster	Higher CAGR; complements Nifty 5
BANKBEES.NS	Banking Sector	Financial Sector Alpha	OVERLAPS: 82% corr with Nifty; 0
MID150BEES.NS	Mid-Cap 150 (NEW)	Alpha & Factor Premium	REPLACES BANK: Real midcap diver
MON100.NS	US Nasdaq 100 (NEW)	Global Tech + Currency Hedge	PROTECTS INR: Low correlation to
GOLDBEES.NS	Precious Metal	Crisis Ballast & Safe Haven	Zero correlation to stocks; prov
SILVERBEES.NS	Precious Metal (NEW)	Industrial Commodity Beta	High volatility commodity upside
GSEC10IETF.NS	10-Yr Sovereign (NEW)	Long-Duration Bond Yield	REPLACES LIQUID: 7.0% yield + ca
LIQUIDBEES.NS	Cash Equivalent	Dry Powder & Rebalance Pool	Zero vol, but acts as a long-ter

HISTORICAL CRISIS STRESS SCENARIOS

Quantifying peak-to-trough drawdowns & multi-asset cushioning

COVID-19 LIQUIDITY CRASH	Feb 20, 2020 - Apr 15, 2020
Nifty 50: -38.4% GoldBees: +12.1%	
Validates Gold as an uncorrelated crisis anchor.	
GLOBAL INFLATION & RATE HIKES	Jan 03, 2022 - Oct 21, 2022
US Tech: -33.1% Nifty: -4.2% USD/INR: +10.2%	
Proves currency depreciation buffers global tech ETF.	
2024 ELECTION & BUDGET SHOCK	Jun 03, 2024 - Jun 05, 2024
Nifty 50: -5.9% single-day shock	
Tests rebalanced G-Sec & Gold stability cushion.	

TIME SERIES & RISK ECONOMETRICS

Advanced statistical indicators integrated into the v2.0 engine

- Rolling Sharpe & Sortino (252d)
Detects risk-adjusted decay and regime change over time.
- EWMA Dynamic Volatility
Decay lambda=0.94 models volatility clustering (ARCH effect).
- Walk-Forward Rebalancing
Tests if quarterly rebalancing generates positive rebalancing alpha.
- Conditional VaR (CVaR 95%)
Quantifies expected shortfall during the worst 5% tail events.
- Maximum Drawdown Duration
Measures recovery time in trading days from peak to new high.

END-TO-END QUANT PRODUCTION PIPELINE

Modular, automated, zero-lookahead-bias architecture from NSE ingestion to serverless API

1. INGEST & HEAL

yfinance download
Split artifact healer
Trading calendar align

2. TIME SERIES ENGINE

Rolling 252d metrics
EWMA vol clustering
Drawdown duration map

3. MPT OPTIMIZER

SLSQP constrained solver
Dirichlet 10k Monte Carlo
Max Sharpe & Min Vol

4. STRESS & BACKTEST

Walk-forward rebalance
COVID / 2022 / 2024 shock
CVaR & Tail risk metrics

5. REST API & REPORT

Vercel serverless Python
Interactive Compare UI
Institutional 12-page PDF